rag -- 2026-11-13

5 new paper(s) · topic: Retrieval-augmented generation: retrieval quality, grounding and faithfulness, evaluation benchmarks, and RAG system design.

Worth reading in full

[P2] CURATOR: Scalable Fine-grained Evidence Selection for Retrieval-Augmented Generation using Decision Models -- P2 introduces a scalable, fine-grained evidence selection framework that dramatically boosts recall and RAG quality while remaining computationally efficient, making it the most impactful advance for practitioners focused on retrieval quality and system design.

Themes

P1, P2, P3 and P5 all grapple with the core tension between retrieval effectiveness and downstream generation quality in RAG systems. The RMIT-ADM+S brief-hybrid configuration shows that modest reductions in search count (12.7 vs ~ 15 searches per topic) can preserve vital coverage (0.892) while cutting cost, yet human judges still prefer a cheaper open-weight model, exposing a gap between automated metrics and perceived answer quality (P1). CURATOR tackles retrieval at the span level, fine-tuning a joint decision model to score up to 128 candidates in a single pass; this yields 17-33 % higher evidence recall and a 21.7× speedup over separate rerankers, directly linking finer-grained retrieval to downstream answer improvements (P2). QDRAG addresses faithfulness by decomposing multi-hop queries and reranking contexts, reporting gains over strong baselines such as Self-Ask and Gemini-Pro on four multimodal QA benchmarks, highlighting that query-level reasoning can mitigate over-citation and improve citation placement (P3). Finally, P5 reveals that even when citations are attached, a substantial fraction (up to ~16 %) of claims stem from the model's parametric memory rather than retrieved passages, and standard NLI-based verification fails (AUROC~0.5), underscoring the need for better provenance detection mechanisms (P5).

Themes cite no papers, so they could not be checked.

At a glance

[P1] RMIT-ADM+S at TREC RAG 2026: Research Agents, Retriever Ablations, and Open-Weight Replication · relevance 0.49
The RMIT-ADM+S team's RAG 2026 runs, especially the brief-hybrid configuration, achieved top automated coverage (0.892 vital) and low cost, but human preferences favored a cheaper open-weight model, highlighting a gap between metrics and perceived quality.

[P2] CURATOR: Scalable Fine-grained Evidence Selection for Retrieval-Augmented Generation using Decision Models · relevance 0.46
CURATOR fine-tunes a pretrained joint decision model to jointly score many candidate evidence spans, achieving higher evidence recall and downstream RAG quality while remaining far faster than separate scoring.

[P3] Faithful in Steps: Improving Generalization and Citation in RAG via Query Decomposition · relevance 0.42
QDRAG improves multi-hop, multi-modal retrieval-augmented generation by decomposing questions, reranking contexts, and finetuning on the decomposition task, achieving stronger answer fidelity and citation quality than prior RAGQ baselines.

[P4] AstroMemBench: A Benchmark for Memory Systems in Embodied Spacecraft Agents · relevance 0.39
AstroMemBench is a new benchmark that evaluates memory systems for embodied spacecraft agents across 16 sub-challenges, showing modest gains for several memory-augmented configurations but no clear superiority over a retrieval-augmented baseline.

[P5] Cited but Unused: Parametric Provenance in Retrieval-Augmented Generation · relevance 0.37
Retrieval-augmented generation often cites sources it didn't actually use, with up to ~16% of claims coming from the model's own memory rather than retrieved passages.

[P1] RMIT-ADM+S at TREC RAG 2026: Research Agents, Retriever Ablations, and Open-Weight Replication

Oleg Zendel, Kun Ran, Chenglong Ma, Nguyen Khoi Vo et al. · arXiv 2611.16425v1 · https://arxiv.org/abs/2611.16425v1 · relevance 0.49

TL;DR: The RMIT-ADM+S team's RAG 2026 runs, especially the brief-hybrid configuration, achieved top automated coverage (0.892 vital) and low cost, but human preferences favored a cheaper open-weight model, highlighting a gap between metrics and perceived quality.

Problem
The study addresses the challenge of building effective research-agent pipelines for Retrieval-Augmented Generation (RAG) on the large-scale ClimbMix corpus, seeking to understand how retrieval strategies, writer models, and workflow designs impact answer quality and cost [1][2].

Method
Five submissions share a staged-evidence research architecture, use adjacent-chunk expansion, and enforce a 500 k-token context budget and a maximum of three citations per sentence [2][3].
Pre-submission ablations used pointwise grading (~31 criteria) and pairwise arena comparisons with GPT-5.6-Terra judges [4][5].

Key results
- Six of seven runs exceeded the track median on all rubric/nugget collections, and the brief-hybrid run tied the track maximum vital coverage of 0.892 for Batch-1 [1].
- Brief-hybrid achieved a higher pointwise quality score (2.333) than the baseline brief-base (2.267) and used 14.4 % fewer searches and 8.2 % fewer processed tokens than brief-base [5].

- Internal pairwise results showed sol-research-first narrowly beating sol-default (67 wins, 51 losses, 1 tie) and brief-hybrid nearly tied brief-base (58 wins, 57 losses, 4 ties) [6].
owa-glm5 required fewer searches per topic (12.7), cost $0.524 per topic versus $2.823 for sol-research-first, had lower automated coverage, and was preferred by human judges on 13 of 17 Batch-1 topics (p = 0.006) [1][7].
- Readability analysis showed owa-glm5 reports were shorter (805 vs. 951 words) and easier to read (Flesch-Kincaid 15.7 vs. 22.2) than sol-research-first, though readability did not correlate with preference margins [7].

Limitations
Pre-submission automated protocols (pointwise grading, pairwise arena) used different assessment methods and topic splits, making their scores not directly comparable to the official NIST results and sensitive to judge and presentation order biases [4][5]. The internal arena only moderately agreed with official coverage rankings (Kendall's ? = 0.33-0.60) and could not reliably reproduce relative system performance, underscoring the limitations of relying solely on automated evaluation [6].

Sources

[1] arXiv 2611.16425v1.pdf, arXiv abstract

[2] arXiv 2611.16425v1.pdf, page 1

[3] arXiv 2611.16425v1.pdf, page 3

[4] arXiv 2611.16425v1.pdf, page 4

[5] arXiv 2611.16425v1.pdf, page 4

[6] arXiv 2611.16425v1.pdf, page 5

[7] arXiv 2611.16425v1.pdf, page 7

[P2] CURATOR: Scalable Fine-grained Evidence Selection for Retrieval-Augmented Generation using Decision Models

Shib Dasgupta, Bandish Parikh, Rashmi Gangadharaiah · arXiv 2611.16366v1 · https://arxiv.org/abs/2611.16366v1 · relevance 0.46

TL;DR: CURATOR fine-tunes a pretrained joint decision model to jointly score many candidate evidence spans, achieving higher evidence recall and downstream RAG quality while remaining far faster than separate scoring.

Problem
Retrieval-augmented generation (RAG) receives whole passages even though only a few spans actually support the answer, and selecting those spans under a limited token budget is essential for effective generation [2]. Existing joint decision models lose discrimination as the number of simultaneous span decisions grows, making them unreliable for large candidate sets [1].

Method
CURATOR adapts the pretrained "pplx-decider-flash-v1.2" model (based on Qwen3.5-4B) by low-rank (LoRA) updates to its attention projections (rank 16, scaling 32) and decision readout, fine-tuning it on variable-size candidate pools (K = 8, 16, 32, 64, 128) with three sampling strategies (natural, fixed-positive, gold-included) to expose the model to both sparse and dense evidence contexts [3][5]. Evidence scores are derived from the difference of positive and negative logits (ri = l(1)i - l(0)i) and used to rank spans within a shared inference pass; the top-scoring spans are greedily admitted until the token budget is exhausted [3].

Key results
- On the QAMPARI training set, CURATOR improves evidence recall by an average 25.8 % over the pretrained joint decider across token budgets [2].
- At a 128-token budget on unseen datasets, CURATOR raises evidence recall by 17.0 % on average and downstream RAG answer quality by 17.4 % relative to the strongest baselines [2].
- In the QAMPARI development set, CURATOR achieves 17-33 % higher recall than the original joint decider and 9-16 % higher recall than separate context-aware scoring across three budgets, while being 21.7× faster and using 25.7× fewer scoring-input tokens than separate scoring [5].
- At a 1024-token budget, CURATOR's recall exceeds span-level Qwen3-Reranker by 64.8 % and BGE-Reranker by 45.5 % [5].
- AUROC for evidence discrimination improves with candidate count, reaching 0.9541 at K = 256 after 2000 training updates, surpassing the original joint decider's 0.6833 at the same K [8].
- General decision-making abilities are preserved: macro-average accuracy rises from 0.8733 to 0.8810 across seven benchmarks after adaptation [8].

Limitations
The paper does not provide quantitative analysis of CURATOR's performance on truly massive candidate pools beyond K = 256, nor does it evaluate multi-step or agentic settings, which are left to future work [6]. Separate context-aware scoring achieves the highest recall on both small diagnostic panels, indicating that access to other candidate evidence can improve individual judgments [7].

Sources

[1] arXiv 2611.16366v1.pdf, arXiv abstract

[2] arXiv 2611.16366v1.pdf, page 3

[3] arXiv 2611.16366v1.pdf, page 5

[5] arXiv 2611.16366v1.pdf, page 7

[6] arXiv 2611.16366v1.pdf, page 10

[7] arXiv 2611.16366v1.pdf, page 12

[8] arXiv 2611.16366v1.pdf, page 17

[P3] Faithful in Steps: Improving Generalization and Citation in RAG via Query Decomposition

Yue Liu, Zhongying Ru, Shimin Di, Jipeng Zhang et al. · arXiv 2611.15538v1 · https://arxiv.org/abs/2611.15538v1 · relevance 0.42

TL;DR: QDRAG improves multi-hop, multi-modal retrieval-augmented generation by decomposing questions, reranking contexts, and finetuning on the decomposition task, achieving stronger answer fidelity and citation quality than prior RAGQ baselines.

Problem
Existing retrieval-augmented generation with quotes (RAGQ) struggles with multi-hop reasoning and multi-modal inputs, leading to over-citation, missed implicit entities, and poor generalization [1][2].

Method
QDRAG introduces three modules: a query decomposer (QD) that breaks a question into atomic subquestions, a context reranker that prunes distractors based on subquestion relevance, and a generator that produces answers with faithful citations [2][3]. Two zero-shot decomposition approaches are proposed: QD-C (entity-replacement + open decomposition) and QD-R (retrieval-reward-driven preference optimization) [2][3]. The system is finetuned on the QD task rather than directly on the downstream RAGQ task, leveraging fast training and better generalizability [1][2][4]. Experiments use 7B and 14B backbones (Qwen2.5-VL-7B, InternVL3-14B) with LoRA adaptation and top-5 context reranking [5].

Key results
- Evaluations are performed on four multi-modal QA benchmarks: MMQA (1,196 queries), WebQA (4,967 queries), InfoSeek, and E-VQA [4].
- Baselines include Self-Ask, SearChain, rewriting models, and instruction-finetuned RAGQs [5].
- QDRAG "significantly outperforms all the baselines on both in-domain and out-of-domain tests, even surpassing Gemini-Pro" [1].
- Finetuning on the QD task yields better generalizability than finetuning on the downstream RAGQ task [1][2].
- Retrieval-feedback annotation is markedly more efficient than LLM-feedback [6].
- Ablation studies show that both the replacement (R) and decomposition (D) operators are critical, and the adaptive context-number assignment module (S) adds performance gains [6].
- QD-R reduces failure cases caused by incorrect context inputs, highlighting the importance of QD-based reranking [6].
The excerpts do not provide specific numeric scores or percentage gains.

Limitations
The provided excerpts do not detail explicit limitations of the QDRAG approach.

Sources

[1] arXiv 2611.15538v1.pdf, arXiv abstract

[2] arXiv 2611.15538v1.pdf, page 2

[3] arXiv 2611.15538v1.pdf, page 3

[4] arXiv 2611.15538v1.pdf, page 4

[5] arXiv 2611.15538v1.pdf, page 5

[6] arXiv 2611.15538v1.pdf, page 6

[P4] AstroMemBench: A Benchmark for Memory Systems in Embodied Spacecraft Agents

You Wu, Guhong Zhang, Hongwei Li, Xinyue Hong et al. · arXiv 2611.16590v1 · https://arxiv.org/abs/2611.16590v1 · relevance 0.39

TL;DR: AstroMemBench is a new benchmark that evaluates memory systems for embodied spacecraft agents across 16 sub-challenges, showing modest gains for several memory-augmented configurations but no clear superiority over a retrieval-augmented baseline.

Problem
Long-term spacecraft autonomy demands agents that can exploit operational history despite communication delays, time-critical operations, environmental uncertainty, physical interaction complexity, power and compute limits, and safety requirements [1]. Existing benchmarks do not fully capture these space-specific memory needs, motivating a dedicated evaluation framework.

Method
The authors introduce AstroMemBench, built with a Scenario-Challenge-Capability design that maps nine planetary-surface, on-orbit servicing, and deep-space tasks into 140 templates and 2 240 executable instances covering 16 memory sub-challenges [1][3]. They evaluate six configurations in a common environment: native Pi Agent (baseline) and five augmentations (RAG, Mem0, Chain-of-Memory, LightMem, and SpaceMind) using the same reasoning model (GLM-5.3-Flash) while varying history length (128 K vs 262 K tokens) and session type (same-session vs new-session) [3][4].

Key results
- Same-session mean scores: Pi 78.20, Pi+RAG 82.74, Mem0 81.70, CoM 82.07, LightMem 79.59; the memory-augmented systems gain 1.39-3.87 points over Pi but remain below RAG [5].
- Token usage rises 13.1%-46.3% for the memory systems, yielding efficiencies of 20.51-25.86 points per billion tokens, lower than Pi's 28.73 and RAG's 26.78 [5].
- New-session scores after a restart: Pi 21.03, RAG 53.03, Mem0 65.28, CoM 52.32, LightMem 30.00, SpaceMind 20.49; Mem0 outperforms RAG by 12.25 points (95 % CI [10.09, 14.37]) [5].
- In the same session, Mem0 scores 1.07 points below RAG (95 % CI [-2.44,+0.34]), while after restart Mem0 leads RAG by 16.28 points (95 % CI [14.32, 18.26]) [5].
- Routine background histories reduce Pi's same-session score by 8.97 points, and native compaction at 128 K tokens improves Pi by 3.54 points compared with a full 262 K history [5].

Limitations
The benchmark shows no clear advantage of the three memory systems (Mem0, CoM, LightMem) over the retrieval-augmented baseline, and the observed benefits mainly support instruction retention and result reuse rather than ongoing state tracking or energy reservation [1][5]. AstroMemBench provides deterministic scoring and the authors plan to refine the benchmark for long-term continuous tasks and richer multimodal interactions in future versions [6].

Sources

[1] arXiv 2611.16590v1.pdf, arXiv abstract

[3] arXiv 2611.16590v1.pdf, page 8

[4] arXiv 2611.16590v1.pdf, page 8

[5] arXiv 2611.16590v1.pdf, page 12

[6] arXiv 2611.16590v1.pdf, page 15

[P5] Cited but Unused: Parametric Provenance in Retrieval-Augmented Generation

Tanzeel Abbas, Tehseen Zia, David Windridge · arXiv 2611.17161v1 · https://arxiv.org/abs/2611.17161v1 · relevance 0.37

TL;DR: Retrieval-augmented generation often cites sources it didn't actually use, with up to ~16% of claims coming from the model's own memory rather than retrieved passages.

Problem
Retrieval-augmented generation systems attach citations to justify claims, but these citations can misstate provenance because the model may rely on parametric memory instead of the retrieved context, and standard verification methods cannot detect this misattribution[1].

Method
The authors define provenance operationally via counterfactual value substitution--replacing a factual value in retrieved passages with a false variant and checking whether the model's regenerated answer changes--allowing automatic labeling of claims as contextual or parametric without human intent judgments[2]. They evaluate five sizes of Qwen2.5-Instruct (0.5B-14B) and Llama-3.1-8B-Instruct on ASQA, ELI5, and QAMPARI benchmarks, using greedy decoding and a DeBERTa NLI verifier as a support-based baseline[2]. Detection experiments compare support-based verification and the model's own distributional signal, measuring AUROC and other metrics[3][4].

Key results
Parametric provenance rates rise with model size: Qwen2.5-0.5B 1.2 % (95 % CI 0.6-2.6), 1.5B 6.5 % (4.5-9.3), 3B 8.3 % (6.1-11.2), 7B 8.2 % (6.0-11.2), 14B 13.3 % (10.4-16.9); Llama-3.1-8B shows 15.7 % (12.6-19.4)[2]. At the most conservative substitution distance the figures are 4.6 % and 8.5 % on the two tested models[1]. Support-based NLI verification performs at chance (AUROC ~ 0.5) across models, even when restricted to the cited passage[3]. The model-distribution signal yields an AUROC of 0.760 with oracle span selection, dropping to 0.668 when scoring all numeric spans, still above chance[4]. Among claims the model could produce without context, 13.2 % were parametric for Qwen2.5-7B[4]. The proposed localised detector costs two forward passes per claim, substantially cheaper than prior attribution methods[8].

Limitations
The study only covers open-weight models from 0.5B to 14B parameters; the method requires token-level log-probabilities with and without context, which closed-source frontier models do not provide[6]. Detection accuracy is not yet sufficient for deployment, as it fails for 1.2% to 15.7% of citable claims and is reported as not currently accurate enough for that use [5][7]. The benchmarks contain no personal data beyond what is already published in their sources [7].

Sources

[1] arXiv 2611.17161v1.pdf, arXiv abstract

[2] arXiv 2611.17161v1.pdf, page 3

[3] arXiv 2611.17161v1.pdf, page 5

[4] arXiv 2611.17161v1.pdf, page 7

[5] arXiv 2611.17161v1.pdf, page 8

[6] arXiv 2611.17161v1.pdf, page 8

[7] arXiv 2611.17161v1.pdf, page 9

[8] arXiv 2611.17161v1.pdf, page 16